

Payment structures in mining software: comparison and analysis

Vendors rarely publish prices, and terms change often, so treat the specifics below as general industry patterns to verify in negotiation.

1. The six models

- A. Perpetual license + annual maintenance.** You pay a large upfront fee to own the right to use a specific version indefinitely. An annual fee (typically 15-22% of the license price) covers support and upgrades.
- B. Subscription (per user / per module).** You pay an annual or monthly fee per named user, concurrent seat, or module, and access stops if you stop paying. Deswik and Seequent Leapfrog work this way.
- C. Cloud/SaaS subscription.** This is like B, but the vendor hosts the software, so there is no on-premise installation. Updates are continuous, and the price often includes infrastructure. Examples are SAP RISE, Maximo SaaS, and cloud tiers of Leapfrog and Datamine.
- D. Metric-based (usage-driven) pricing.** The price scales with a measured quantity: data tags for historians like PI, asset counts for Maximo, tonnes, or number of machines. It is often layered on top of A, B, or C.
- E. Project-based hybrid (hardware + implementation + recurring).** You pay upfront for hardware, installation, customization, and training, then a recurring license and service fee. This is typical of fleet management (DISPATCH, Wenco) and integrated operational systems.
- F. Enterprise license agreement (ELA).** A multi-year, multi-site negotiated contract, usually a fixed or capped annual fee covering a defined product set. Large miners typically use it as a wrapper around the models above.

2. Side-by-side comparison

Dimension	A. Perpetual + maintenance	B. On-prem subscription	C. Cloud/SaaS	D. Metric-based	E. Project hybrid	F. Enterprise agreement
Upfront cost	Very high	Low	Lowest	Low to medium	Very high	Medium (often negotiated)

Dimension	A. Perpetual + maintenance	B. On-prem subscription	C. Cloud/SaaS	D. Metric-based	E. Project hybrid	F. Enterprise agreement
Ongoing cost	Moderate (maintenance)	High	High	Grows with usage	Moderate	Fixed or capped
Cost over 7-10+ years	Usually lowest	Highest	Highest, though it includes IT costs	Unpredictable	Moderate to high	Lower per unit at scale
Budget treatment	CapEx	OpEx	OpEx	OpEx	Mixed	Mixed
Cost predictability	High	High (until renewal)	Medium (price escalators)	Low	Medium	Highest
Flexibility to scale down	Poor (sunk cost)	Excellent	Excellent	Excellent	Poor	Limited by commitment
Access to latest features	Only with active maintenance	Always	Always, automatic	Depends on base model	Depends on contract	Usually included
Vendor lock-in	Medium	High	Highest	High	Highest	High
IT burden on you	High	High	Low	Depends	High	Depends
Works offline / remote sites	Yes	Mostly	Poorly	Depends	Yes (usually on-site)	Depends
Best for	Stable long-life mines	Project-based teams	Collaboration, small-to-mid teams	Variable-scale operations	Mission-critical operations	Large multi-site miners

3. Benefits and risks of each

A. Perpetual + maintenance

Benefits

- Lowest total cost over a long horizon, since maintenance is only a fraction of the license fee.

- You can keep running a frozen version if you stop paying, which is valuable for validated, audited resource models.
- Predictable costs, and works without internet.
- The asset can be capitalized, which suits some balance-sheet strategies.

Risks

- Large upfront outlay in a cyclical industry.
- You pay for licenses you may not use during care-and-maintenance periods.
- Lapsed maintenance can mean expensive reinstatement or upgrade fees.
- Technology can be left behind, and vendors are actively steering customers away from this model.

B. On-premise subscription

Benefits

- Low entry cost, so a small team or a new project can start immediately.
- Scale seats up or down with project phases (exploration, feasibility, production).
- Always on the current version, and lower obsolescence risk.

Risks

- Cumulative cost becomes the highest over a long mine life.
- Access is cut off if payment stops, which is a risk for historical data and models.
- Renewal price increases are hard to resist once you're embedded.

C. Cloud/SaaS

Benefits

- Minimal IT overhead: no servers, patching, or version management.
- Real-time collaboration between site, head office, and consultants.
- Fast deployment across multiple sites, and elastic compute for heavy modeling.

Risks

- Remote mine sites often have unreliable connectivity, which can make cloud tools impractical for operations.
- Data sovereignty and regulatory constraints in some jurisdictions.
- Deepest lock-in, since data export and exit can be difficult.
- Cybersecurity concerns where systems touch operational technology.
- Under IFRS, some SaaS implementation and customization costs must be expensed rather than capitalized, which can hurt reported earnings. Check with your accountants.

D. Metric-based pricing

Benefits

- Cost follows actual use, so you don't pay for capacity you don't need.
- Fair for operations that ramp up or wind down.

Risks

- Budget surprises when operations expand or when sensor and data volumes grow.
- Metrics can be defined in the vendor's favor, so scrutinize the definitions (what counts as a "tag" or "asset").
- It discourages using the software more, which is the opposite of what you want.

E. Project-based hybrid

Benefits

- Bundled hardware, software, and integration from one accountable vendor.
- Deep customization to the site's fleet, layout, and processes.
- The vendor is contractually on the hook for uptime and performance.

Risks

- High switching costs, because hardware and software are tied together.
- Implementation costs can equal or exceed the software fees, and overruns are common.
- Vendor-locked ecosystems (for example, a fleet management system that works best with the same OEM's machines).

F. Enterprise agreement

Benefits

- Volume discounts and price caps on renewals.
- One negotiation instead of many, and simpler compliance and audit.
- Flexibility to swap between products in the suite, and dedicated vendor support.

Risks

- Overcommitment: you can end up paying for shelfware.
- Long terms (3-5 years) are risky if a mine closes or is sold, so negotiate assignment and downsizing clauses.
- Needs strong internal license management to capture the value.

4. Illustrative break-even example

These numbers are assumptions to show the logic, not real quotes.

- **Perpetual:** 100k*license* + 2020k/yr)
- **Subscription:** \$35k/yr

Year	Perpetual (cumulative)	Subscription (cumulative)
1	\$120k	\$35k
3	\$160k	\$105k
5	\$200k	\$175k
7	\$240k	\$245k
10	\$300k	\$350k

The break-even is about year 7. If the tool is needed for 3-4 years (a short project), subscription wins clearly. For a 15+ year mine, perpetual wins by a wide margin, provided the vendor still offers it and you stay on maintenance. The analysis changes further when you add the IT and hosting costs of on-premise (which favor subscription or SaaS) and discount future cash flows (which favors deferring cost).

5. Mining-specific factors that shift the decision

1. **Mine life and lifecycle stage.** Exploration and feasibility teams favor subscription flexibility, while long-life producing assets favor perpetual or enterprise agreements.
2. **Commodity cycles.** Low OpEx commitments are safer when prices crash, and subscriptions can be cut. Sunk CapEx cannot.
3. **Connectivity.** Remote sites push operational systems (fleet management, historians) toward on-premise or hybrid, and planning tools toward cloud or subscription at head office.
4. **Regulatory and audit needs.** Reserve and resource reporting (JORC, NI 43-101, SAMREC) requires reproducible models, and the ability to reopen old models on frozen software is a real advantage of perpetual licensing.
5. **Operational criticality.** For systems where downtime stops production (dispatch, control), the vendor's service commitments matter more than the license price.
6. **Ecosystem lock-in.** Pairing OEM equipment with the same OEM's software may be commercially attractive but reduces future negotiating leverage.
7. **Accounting.** CapEx versus OpEx treatment affects EBITDA, covenants, and tax, so involve finance early.

6. How this maps to the ten tools

Software	Most likely fit	Why
Hexagon MinePlan	Enterprise agreement or perpetual → subscription	Core long-term planning tool for large miners

Software	Most likely fit	Why
Deswik	Subscription	Modular, seat-based, scales with team
Maptek Vulcan	Perpetual or lease	Long-standing core modeling tool
Datamine	Perpetual or lease, cloud emerging	Long mine lives, modeling continuity
Leapfrog	Subscription / cloud	Collaboration-heavy, project-based use
DISPATCH	Project hybrid	Hardware-tied, mission-critical
Wenco	Project hybrid	Same
AVEVA PI	Metric-based subscription	Priced on data tags
SAP	Cloud subscription + heavy implementation	Enterprise-wide, long lifecycle
Maximo	SaaS subscription, metric-based	Priced by users and assets

7. Practical recommendations

- **Match the model to the time horizon.** Under about 5 years, subscription usually wins. Over about 8 years, perpetual or a fixed-price enterprise agreement usually wins.
- **Use a hybrid portfolio.** Many large miners use perpetual or enterprise agreements for core geology and planning, subscription for specialist and collaborative tools, and hybrid contracts for operational systems.
- **Negotiate the exit, not just the entry.** Secure data export rights, access to historical files, price-increase caps (for example, CPI or 3-5% maximum), and rights to reduce seats or transfer licenses on mine sale or closure.
- **Model total cost of ownership, not license price.** Include implementation, integration, training, IT infrastructure, and internal admin time.
- **Ask for a multi-year quote with escalators.** Year-one pricing is often a teaser.
- **Test the cloud against your worst connectivity day** before committing operational workloads to it.

I can build a TCO calculator or a scoring matrix for your own numbers, or narrow this to a specific tool or company size if you tell me the scenario.

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Short answer: it's mostly not a one-time payment. The industry started with perpetual licenses plus annual maintenance and is shifting to subscriptions. Almost all of it is quote-based, so prices aren't published. The models below come from my general knowledge and vendors change them, so confirm with a vendor for current terms.

#	Software	Used for	Typical payment structure
1	Hexagon MinePlan	Geological modeling, mine planning	Historically perpetual license plus annual maintenance. Subscription and enterprise agreements are now common.
2	Deswik (Deswik.Suite)	Mine design, scheduling	Annual subscription, licensed per user and per module.
3	Maptek Vulcan	Geology, modeling, planning	Perpetual plus annual maintenance, or annual lease or subscription.
4	Datamine (Studio, etc.)	Resource modeling, scheduling	Perpetual plus maintenance, or annual lease. Cloud-based options are growing.
5	Seequent Leapfrog	3D geological modeling	Subscription, per seat, with tiers (Geo, Works, Central for collaboration).
6	Modular Mining (Komatsu) DISPATCH	Fleet management and truck dispatch	Project-based: implementation fees plus recurring licensing, support and service contracts, often bundled with hardware.
7	Wenco (Hitachi)	Fleet management	Similar to DISPATCH: upfront hardware and implementation, then a recurring service or license fee.
8	AVEVA PI System (formerly OSIsoft)	Process data historian	Was perpetual plus maintenance, now mostly subscription, often priced by data tags or points.
9	SAP (S/4HANA, mining solutions)	ERP: finance, procurement, supply chain	Cloud subscription (RISE with SAP), or perpetual plus maintenance for older on-premise setups. Implementation costs are large and separate.
10	IBM Maximo	Asset and maintenance management	Subscription (SaaS or hosted), or perpetual plus support. Priced by user type or asset volume.

Patterns across the industry

- **Subscription** is the direction of travel, especially for planning and modeling tools, because vendors prefer recurring revenue and miners increasingly want cloud access.
- **Perpetual plus maintenance** is still offered by some vendors, with maintenance typically around 15 to 22% of the license fee per year.
- **Hybrid or project-based** pricing is typical for operational systems like fleet management. It includes hardware, installation, customization, training and multi-year service contracts.
- **Enterprise agreements** are common for large miners. They negotiate multi-year, multi-site deals with volume discounts rather than paying list price.
- Implementation, customization, integration and training are usually charged on top, and they can rival the software cost itself.

I can look up current pricing or licensing news for any of these if you like. Are you comparing vendors for a purchase, or researching the market?